# DATA CENTER SIMULATION PERFORMANCE REPORT

Profile: NORMAL_DAY_SAMPLE | Generated: 2026-08-23 16:09:08

| Parameter | Value |
|---|---|
| Experiment Duration | 600.0 seconds |
| Start Time (T1) | 1787480947.3827925 |
| End Time (T2) | 1787481547.3827925 |
| Primary Bottleneck | Application / Endpoint Latency |
| Confidence Level | LOW_SAMPLE_SIZE |
| Failure Classification | SLA_NON_COMPLIANCE |

| SLA Metric | Measured Value | Target Limit | Result |
|---|---|---|---|
| P95 Response Time | 1900.0 ms | 500.0 ms | FAIL |
| Error Rate | 0.0 % | 1.0 % | PASS |
| Peak CPU Utilization | 0.58 % | 80.0 % | PASS |
| Peak Memory Utilization | 17.27 % | 90.0 % | PASS |

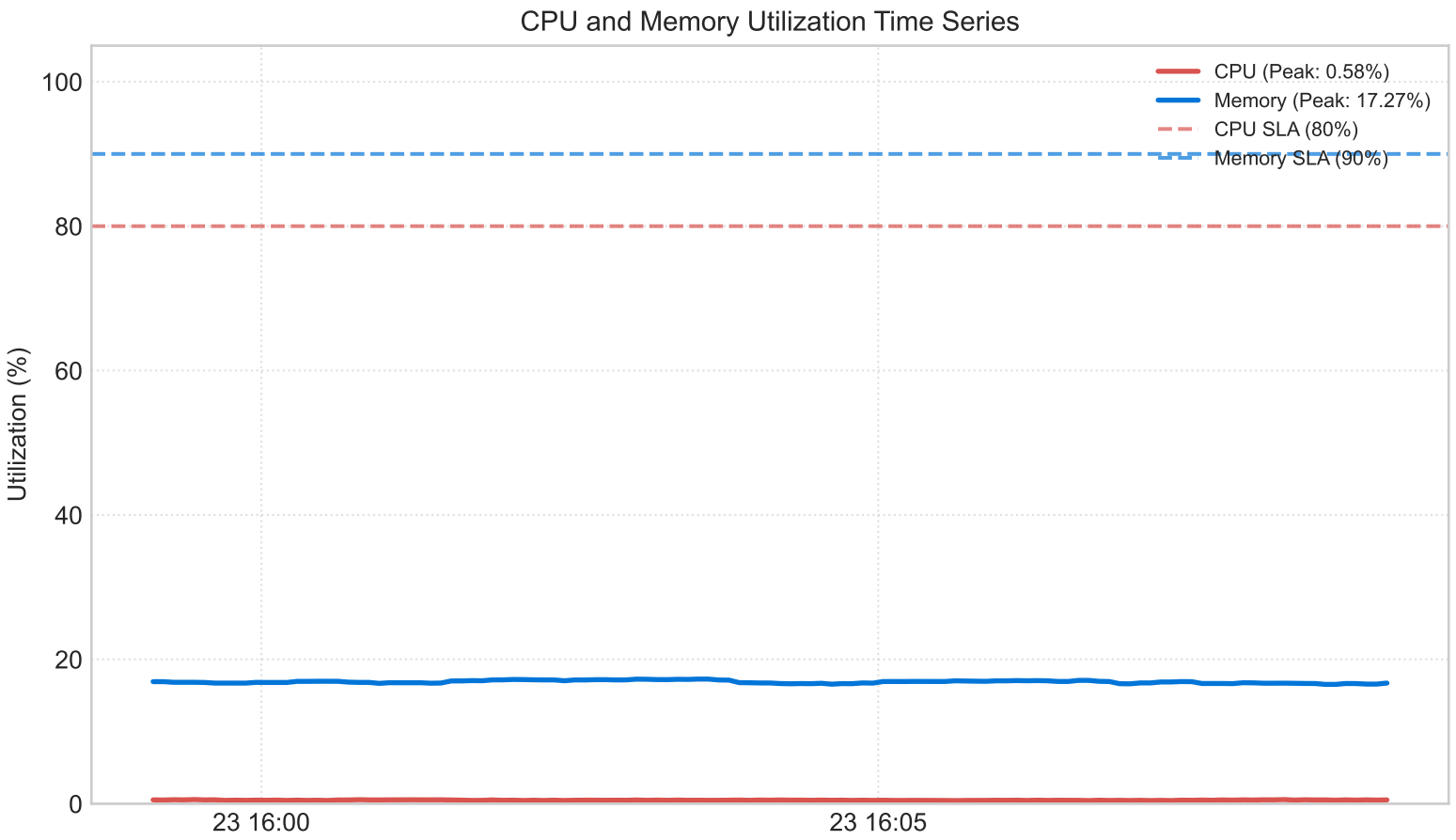
**Executive Conclusion & Evidence Synthesis:**

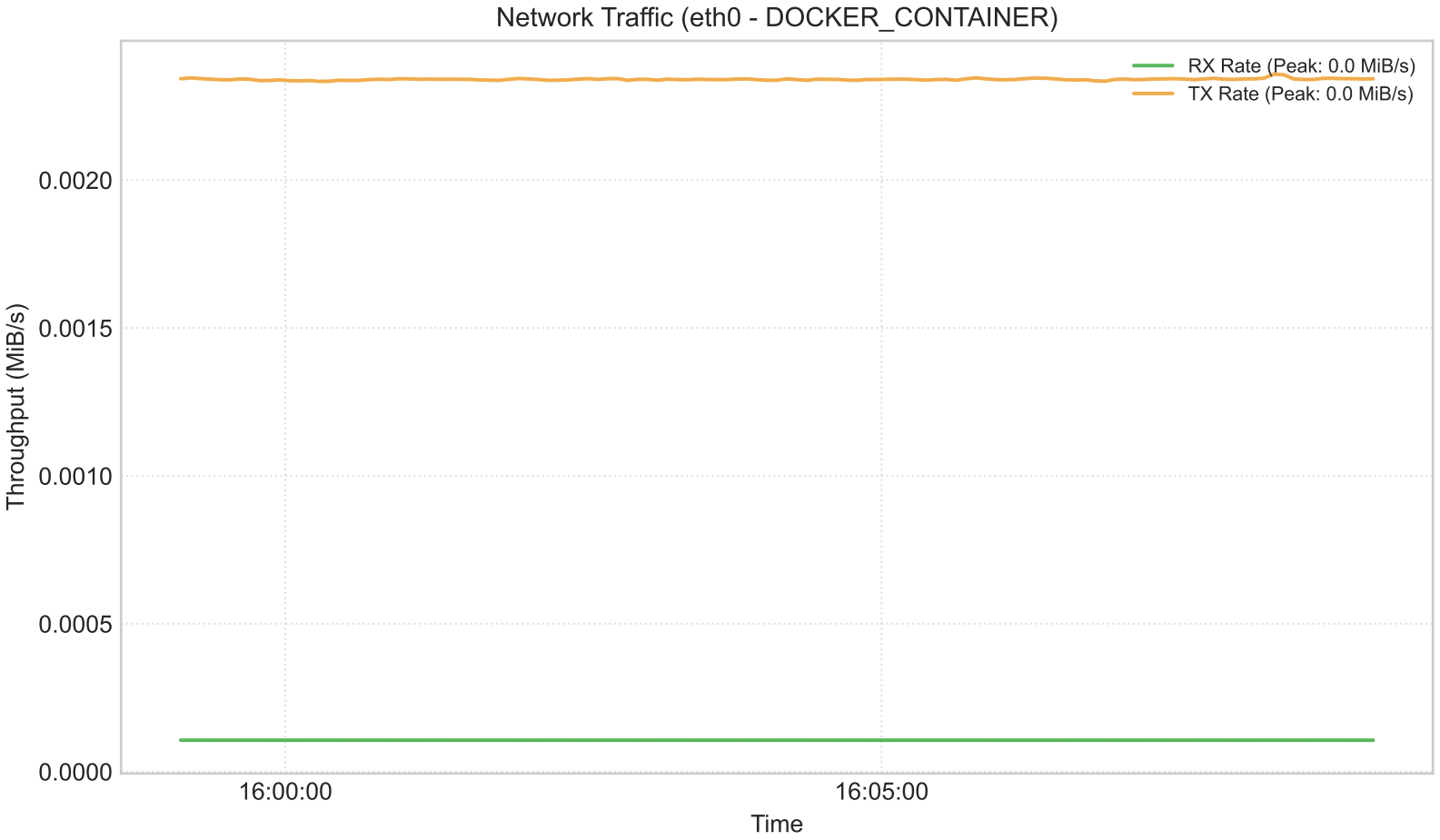
The experiment executed 41 requests with overall SLA status 'FAIL' (3/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. Primary bottleneck assessed as 'Application / Endpoint Latency' (Confidence: LOW_SAMPLE_SIZE).

**Diagnostic Observations:**

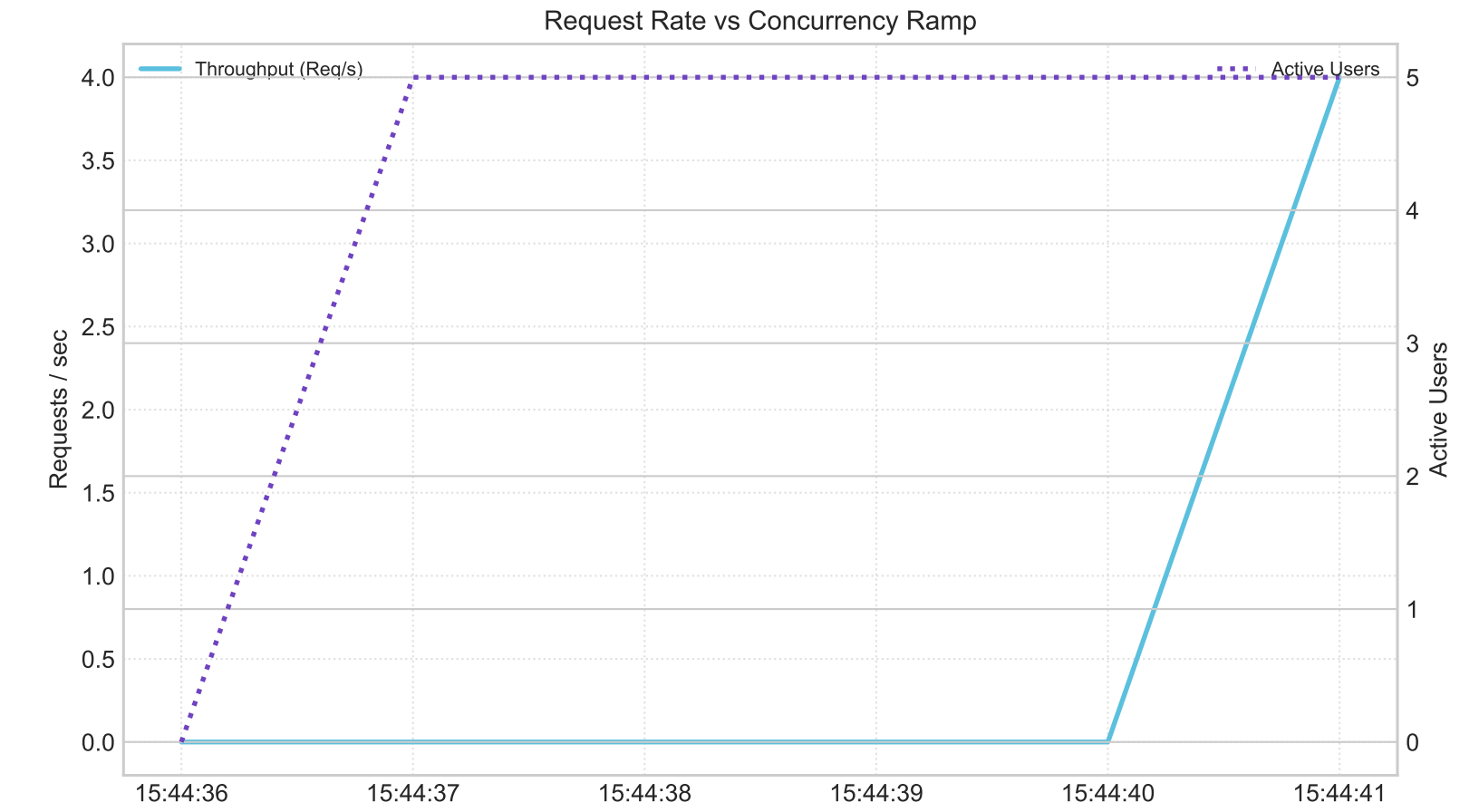
- Low request volume (41 requests); statistical latency metrics are tentative.
- P95 response time (1900.0 ms) violated SLA target (500.0 ms).
- Endpoint bottleneck identified on 'POST /login' with P95 of 1900.0 ms.

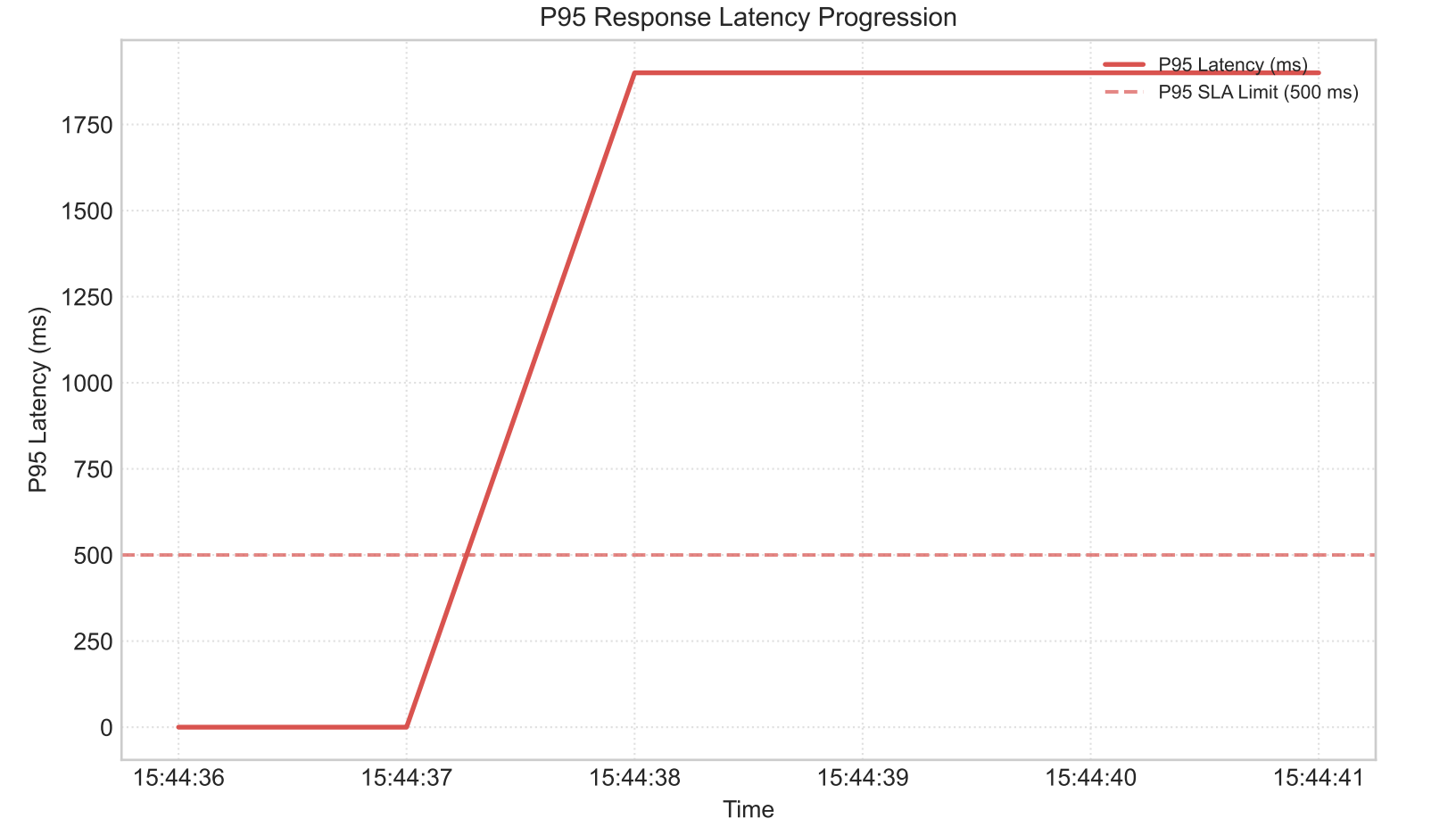
# System & Hardware Resource Dynamics



# Locust Workload & Latency Profile



# Endpoint Breakdown & Chronological Timeline

## HTTP Endpoint Performance Metrics:

| Endpoint | Requests | Fails | Avg Latency | P95 Latency | RPS |
|---|---|---|---|---|---|
| GET /api/data | 14 | 0 | 76.34 ms | 140.0 ms | 1.56 |
| GET /health | 9 | 0 | 10.55 ms | 19.0 ms | 1.0 |
| POST /login | 5 | 0 | 1885.21 ms | 1900.0 ms | 0.56 |
| GET /me | 12 | 0 | 13.19 ms | 28.0 ms | 1.34 |
| POST /profile/update | 1 | 0 | 61.69 ms | 62.0 ms | 0.11 |

## Chronological Event Timeline:

| Time | Event Type | Category | Severity | Description |
|---|---|---|---|---|
| 15:[LOCUST] LOAD_STARTED | | OBSERVED | INFO | Workload generation began with 0 users |
| [LOCUST] 15:[LOCUST_PEAK_USERS | | OBSERVED | INFO | Locust concurrency reached peak of 5 active u... |
| [SLA_ENGINE] 15:[SLA_BREACH_DETECTED | | INFERRED | CRITICAL | SLA Target Breached: P95 Response Time was 19... |
| [LOCUST] 15:[LOCUST_PEAK_RPS | | OBSERVED | INFO | Locust throughput reached peak of 4.0 req/s |
| [RUNNER] 15:[EXPERIMENT_START | | OBSERVED | INFO | Experiment started at (1787480947.3827925) |
| [PROMETHEUS] 15:[PROMETHEUS_CPU_PEAK | | OBSERVED | INFO | CPU utilization peaked at 0.58% |
| [PROMETHEUS] 15:[NETWORK_PEAK_RX | | OBSERVED | INFO | Network receive rate peaked at 0.0 MiB/s on e... |
| [PROMETHEUS] 15:[PROMETHEUS_MEMORY_PEAK | | OBSERVED | INFO | Memory utilization peaked at 17.27% |
| [PROMETHEUS] 15:[NETWORK_PEAK_TX | | OBSERVED | INFO | Network transmit rate peaked at 0.0 MiB/s on ... |
| [RUNNER] 15:[EXPERIMENT_END | | OBSERVED | INFO | Experiment completed at (1787481547.382792... |